Trade Study: Battery

- **Conducted by:** Will Huff
- **Decision:** Two batteries, one for motors, one for the Jetson

Problem: The car will need two different voltage loads. The Jetson requires a steady 9–19V supply that cannot dip, because a voltage drop can corrupt the microSD card and cause the Jetson's components to malfunction. The motors need about 12V and will be very power-hungry; whenever they start up, they will pull the battery voltage down. If both loads are wired to the same battery, it is almost guaranteed that the Jetson's supply will drop below 9V every time the motors start and whenever they demand more power while driving. For this reason, the power system must have two separate paths.

		One Central Power Source	Two Batteries
Criteria	**Weights**	**Grade**	**Grade**
Electrical compatibility (Voltage and amperage)	0.3	5	8
Runtime	0.2	6	8
Cost	0.2	7	6
Build Effort	0.1	5	9
Simplicity	0.2	5	7
Average Score		5.6	7.6
Weighted Score	1.0	5.6	7.5

Breakdown

Option 1: One central power source

Using a single power source would mean one large battery with multiple power lines branching from it. One battery would save space on the surface of the GEEK smart robot, giving us more freedom with other parts of the build. It is also arguably simpler to charge: rather than monitoring the charge levels of two separate power sources, we could just plug the car in when the battery runs low. The major downside is that it would make managing the electrical outputs much more difficult. Because the Jetson draws power based on its workload while the motors need 12V, one battery would require two separate circuits running from a single source. This can be difficult to manage and would require extra components such as a buck converter and fuse holders. A bigger problem is the risk of SD card corruption and component malfunction on the Jetson. If the Jetson's supply ever drops below 9V, it can corrupt the SD card and potentially damage the board. This is likely to happen with one battery, because any time the motors draw extra power, they pull voltage away from the Jetson. While managing one main power source may seem cleaner, it would likely make things much more difficult.

Option 2: Two batteries for each component

Using two batteries, one for the motors and one for the Jetson, would involve a small 12V rechargeable battery made up of three 18650 cells and a compact power bank capable of delivering up to 65W. This setup would take up more space on the chassis than a single battery, and it may cost more, but it is worth it overall. The benefits of two batteries are electrical simplicity and reliable Jetson operation. The power bank provides more than enough wattage for the Jetson during demanding tasks, while the motors receive their 12V from a completely separate source. This approach also requires no additional components beyond a microcontroller connecting the Jetson to the motors. Overall, this build would be much simpler and easier to manage.